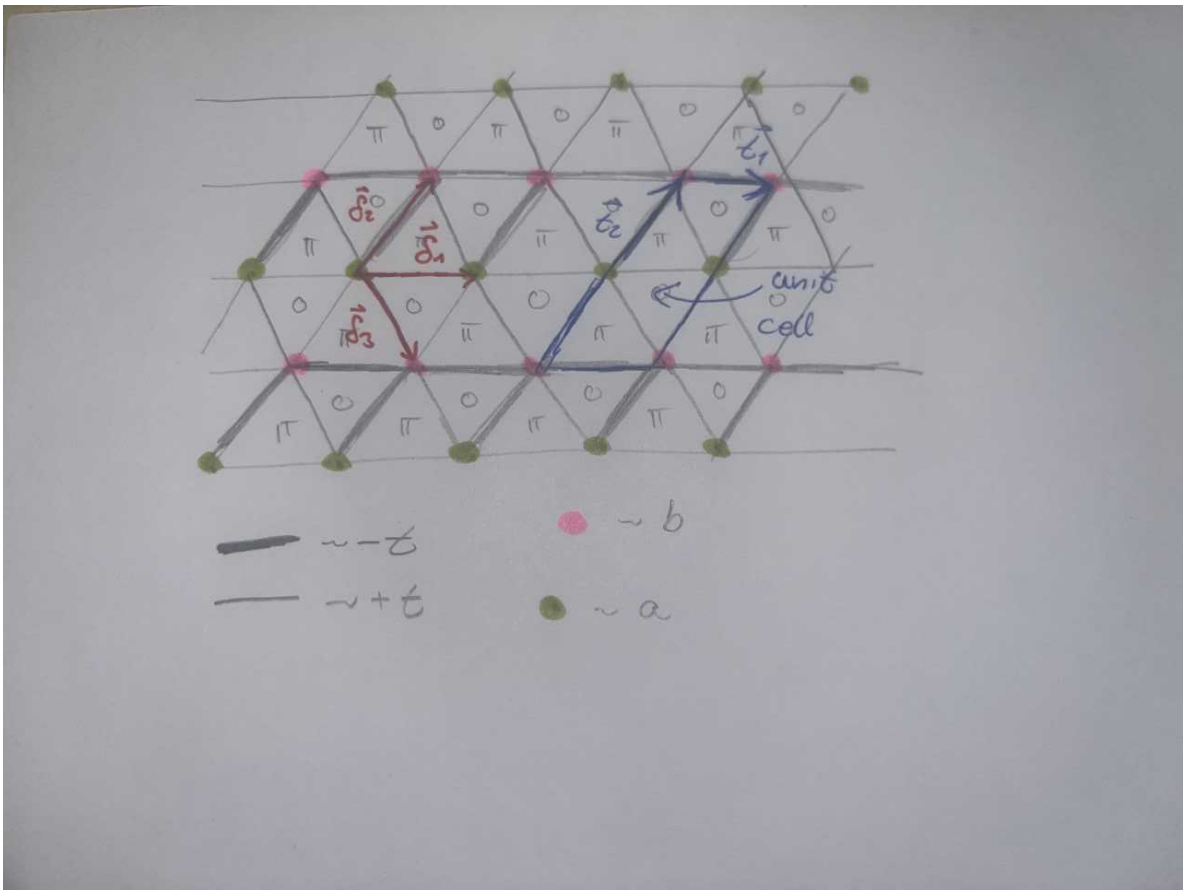
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0.1 0-π and π-0 States

0.1.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitve translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{1}$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij}, and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2} \sum_{\langle i,j \rangle, \sigma} \left(t a^\dagger_{i\sigma} a_{j\sigma} - t b^\dagger_{i\sigma} b_{j\sigma} + t^{(ab)}_{ij} a^\dagger_{i\sigma} b_{j\sigma} + \text{h.c.} \right) + \mu_a \sum_{i,\sigma} \left(a^\dagger_{i\sigma} a_{i\sigma} - 1 \right) + \mu_b \sum_{i,\sigma} \left(b^\dagger_{i\sigma} b_{i\sigma} - 1 \right) + \text{const.} \tag{2}$$

$$\begin{aligned} H = \frac{t}{2} \sum_{i\sigma} \Big(& a^\dagger_{i\sigma} a_{i+\delta_1;\sigma} + a^\dagger_{i\sigma} a_{i-\delta_1;\sigma} - a^\dagger_{i\sigma} b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \\ & - b^\dagger_{i\sigma} b_{i+\delta_1;\sigma} - b^\dagger_{i\sigma} b_{i-\delta_1;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_2;\sigma} - b^\dagger_{i\sigma} a_{i-\delta_2;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_3;\sigma} + b^\dagger_{i\sigma} a^\dagger_{i-\delta_3;\sigma} \\ & + \mu_a a^\dagger_{i\sigma} a_{i\sigma} + \mu_b b^\dagger_{i\sigma} b_{i\sigma} \Big) - \frac{N(\mu_a + \mu_b)}{2} \end{aligned} \tag{3}$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2}\sum_{i,\delta} 2\tilde{J}\left(1-6|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 = 6N\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \tag{4}$$

This is not quite wright yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}}\sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \tag{5}$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma}\begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma}\end{pmatrix}\begin{pmatrix} 2t\cos\left(\vec{q}\vec{\delta}_1\right)+\mu_a & 2t\cos\left(\vec{q}\vec{\delta}_3\right)-2it\sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t\cos\left(\vec{q}\vec{\delta}_3\right)+2it\sin\left(\vec{q}\vec{\delta}_2\right) & -2t\cos\left(\vec{q}\vec{\delta}_1\right)+\mu_b\end{pmatrix}\begin{pmatrix} a_{q\sigma} \\ b_{q\sigma}\end{pmatrix}. \tag{6}$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi\begin{pmatrix}0\\1/\sqrt{3}\end{pmatrix}, \quad \vec{b}_2 = 2\pi\begin{pmatrix}1\\-1/\sqrt{3}\end{pmatrix} \tag{7}$$

.
The suitable transformation is

$$\begin{pmatrix}k_x\\k_y\end{pmatrix} = \begin{pmatrix}1 & \sqrt{3}\\ m1 & 0\end{pmatrix}\begin{pmatrix}q_x\\q_y\end{pmatrix} \tag{8}$$

$$\Leftrightarrow \begin{pmatrix}q_x\\q_y\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1/\sqrt{3} & -1/\sqrt{3}\end{pmatrix}\begin{pmatrix}k_x\\k_y\end{pmatrix} \tag{9}$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \tag{10}$$

This yields

$$\cos\left(\vec{q}\vec{\delta}_1\right) \to \cos(k_y)$$

$$\sin\left(\vec{q}\vec{\delta}_2\right) \to \sin(k_x/2) \tag{11}$$

$$\cos\left(\vec{q}\vec{\delta}_3\right) \to \cos(k_y - k_x/2) \tag{12}$$

$$\varepsilon(\vec{k})_\pm = \mu_a+\mu_b\pm\sqrt{(\mu_a-\mu_b)^2+2t\cos(k_y)(\mu_a-\mu_b)+4t^2\cos^2(k_y)+4t^2\cos^2(k_y-k_x/2)+4t^2\sin^2(k_x/2)} \tag{13}$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-¿ charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_\pm = \pm 2|t|\sqrt{(\mu/(2t)+\cos(k_y))^2+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \tag{14}$$

It is correct that ε only depends on the absolute value of t, it looks like this would completely neutrilise the particle-hole transformation.The periodicity is a bit weird here, since the expression is not periondic in 2π in every direction as one would expect. I am prettys rue this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brilouin zone got cut in hoalf in one direction.

represent hamiltonian as

$$H = \sum_{q,\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} \end{pmatrix} \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \tag{15}$$

$$\vec{h} = \begin{pmatrix} 2t\cos(k_y - k_x/2)) \\ 2t\sin(k_x/2) \\ 2t\cos(k_y) + \mu \end{pmatrix} \tag{16}$$

Then $\varepsilon(\vec{q})_\pm = \pm|\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\cos(k_y) + \mu)}} \begin{pmatrix} \varepsilon_+ + 2t\cos(k_y) + \mu & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\cos(k_y) + \mu \end{pmatrix} \tag{17}$$

is the transformation matrix U. It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a^\dagger_{i\uparrow} b_{j\uparrow} \rangle &\overset{\text{FT}}{=} \frac{2}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a^\dagger_p b_q \rangle}_{\propto \delta_{q,p}} \\ &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a^\dagger_q b_q \rangle \\ &\overset{\text{BT}}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left(u^*_{11} u_{21} f(\varepsilon_+) + u^*_{12} u_{22} f(\varepsilon_-) \right) \end{aligned} \tag{18}$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u^*_{12} = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned} u^*_{11} u_{21} f(\varepsilon_+) + u^*_{12} u_{22} f(\varepsilon_-) &\overset{1.}{=} (u^*_{11} u_{21} - u_{22} u^*_{12}) f(\varepsilon_+) + u_{22} u^*_{12} \\ &\overset{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\ &= u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \tag{19}$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \to ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned} \Rightarrow \langle a^\dagger_{i\uparrow} b_{j\uparrow} \rangle &= \Omega \int\limits_{1.\text{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\ &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \tag{20}$$

$$u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{21}$$

Now for the other expectation values:

$$\Rightarrow \langle b^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] u_{11} u^*_{21} \left(2f(\varepsilon_+) - 1\right) \tag{22}$$

$$\begin{aligned}
\langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \ldots &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)\right] \\
&\overset{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[|u_{12}|^2 - |u_{11}|^2) f(\varepsilon_-) + |u_{11}|^2\right] \\
&\overset{5.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[(1 - 2u^2_{11}) f(\varepsilon_-) + u^2_{11}\right] \\
&\overset{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[(1 - 2u^2_{11})(1 - f(\varepsilon_+)) + u^2_{11}\right]
\end{aligned} \tag{23}$$

$$\Rightarrow \langle a^\dagger_{i\uparrow} a_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] \left[(1 - 2u^2_{11})(1 - f(\varepsilon_+)) + u^2_{11}\right] \tag{24}$$

$$\Rightarrow \langle b^\dagger_{i\uparrow} b_{j\uparrow}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y)\right)^T (\vec{R}_j - \vec{R}_i)\right] \left[(1 - 2u^2_{11}) f(\varepsilon_+) + u^2_{11}\right] \tag{25}$$

$$u^2_{11} = (1/2\varepsilon_+)(\varepsilon_+ + 2t\cos(k_y) + \mu) \tag{26}$$

$$\langle N^a\rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} (1 - 2u^2_{11}) f(\varepsilon_-) + u^2_{11} \tag{27}$$

$$\langle N^b\rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} (1 - 2u^2_{11}) f(\varepsilon_+) + u^2_{11} \tag{28}$$

The constraint $\langle N^a_i\rangle = 1 = \langle N^b_i\rangle$ should, as gloal halffiling is already ensured and $\langle N^a\rangle$ and $\langle N^b\rangle$ are uniform by construction, be equivalent to $\langle N^a\rangle = \langle N^b\rangle$. Using the expressions above:

$$\langle N^a\rangle - \langle N^b\rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} (1 - 2u^2_{11})(1 - 2f(\varepsilon_+)) \overset{!}{=} 0 \tag{29}$$

Numerics say that $\langle N^a\rangle = \langle N^b\rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). I feel like there is a more suphisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1 - 2f(\varepsilon_+)$.

Sanity check:

$$\langle N^a + N^b\rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} 1 = 2 \tag{30}$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\left[(1-2u_{11}^2)(1-f(\varepsilon_+))+u_{11}^2\right] \tag{31}$$

$$\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\left[(1-2u_{11}^2)f(\varepsilon_+)+u_{11}^2\right] \tag{32}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{i(k_y-k_x/2)}u_{11}u_{21}(2f(\varepsilon_+)-1) \tag{33}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_x/2}u_{11}u_{21}^{*}(2f(\varepsilon_+)-1) \tag{34}$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.

Another observation:

$$\begin{aligned}\frac{1}{2}\sum_{\sigma}\left(\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle + \langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle\right) &= \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}e^{ik_y}\\ &= \frac{1}{2\pi}\int\limits_{[-\pi,\pi]}dk_y e^{ik_y}\\ &= \frac{\sin(\pi)}{\pi}\\ &= 0\end{aligned} \tag{35}$$

$$\Rightarrow \frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = -\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle, \tag{36}$$

as expected. The numerical results are

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = 0.19754 \tag{37}$$

$$\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle = -0.19754 \tag{38}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_3\sigma}\rangle = 0.19754 \tag{39}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_2\sigma}\rangle = 0.19754 \tag{40}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i-\delta_2\sigma}\rangle = -0.19754 \tag{41}$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \tag{42}$$

$$\tag{43}$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}\frac{\langle H\rangle}{N} &= \frac{1}{N}\left\langle \sum_{q\sigma} \varepsilon_{+\sigma}\alpha^\dagger_{+\sigma}\alpha_{+\sigma} + \varepsilon_{-\sigma}\alpha^\dagger_{-\sigma}\alpha_{-\sigma}\right\rangle \\ &\overset{\text{th.lim.}}{\longrightarrow} 2\frac{1}{N}\frac{\Omega N}{2}\int\limits_{1.\text{BZ}} \frac{d^2q}{(2\pi)^2}\varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\ &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(2f(\varepsilon_+)-1) \end{aligned} \tag{44}$$

(the blue 2 in the second line is the spin degeneracy)
Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+(2f(\varepsilon_+)-1) \overset{\text{ph-trafo}}{\longrightarrow} \varepsilon_-(2f(\varepsilon_-)-1) = -\varepsilon_+(1-2f(\varepsilon_+)) = \varepsilon_+(2f(\varepsilon_+)-1) \tag{45}$$

The full energy density is:

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) = -1.2206\ J \tag{46}$$

0.1.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa,\chi,T=0)}{N} = \frac{E_0}{N} + \frac{1}{N}\sum_{\vec{k}}\varepsilon(\vec{k}) \tag{47}$$

$$E_0 = 6N\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\right) \tag{48}$$

$$\varepsilon(\vec{k}) = -2\left|-\tilde{J}\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\right|\sqrt{\cos^2(k_1)+\cos^2(k_3)+\sin^2(k_2)} \tag{49}$$

From now on I use the shorthand $|\langle\chi_{ij}\rangle| \equiv \chi$.

$$\begin{aligned}\Rightarrow f_0 &= \tilde{J}\left(6(1-6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma\left|\chi(1-4\tilde{\kappa}\chi^2)\right|\right) \\ &= \tilde{J}\left(\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4\right)\end{aligned} \tag{50}$$

with

$$\gamma = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\cos^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.18524 \tag{51}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates f/M ($f/2$)!

$$\frac{\partial f_0}{\partial\chi} = \tilde{J}\left(\pm 2\gamma - 12\chi\right)\left(12\tilde{\kappa}\chi^2 - 1\right) \qquad \frac{\partial^2 f_0}{\partial\chi^2} = \tilde{J}\left(12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2\right) \tag{52}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \qquad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{53}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6}\left(\frac{\tilde{\kappa}\gamma^2}{18} - 1\right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \tag{54}$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.\bar{6}$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \tag{55}$$

with the free energy

$$f_0 = -1.22206\, J. \tag{56}$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

0.1.3 fermi surface

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t|\sqrt{\cos^2(q_x) + \sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right)} \tag{57}$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right) = 1 + \sin(q_x)\sin\left(\sqrt{3}q_y\right) \tag{58}$$

,
I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = \underbrace{-\sin(q_0^x)\sin\left(\sqrt{3}q_0^y\right)}_{\in[-1,1]}. \tag{59}$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1 + \cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_x^0 = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \tag{60}$$

$$\Rightarrow \sin\left(\sqrt{3}q_0^y\right) = \begin{cases} -1 & \text{k even} \\ 1 & \text{k odd} \end{cases} \tag{61}$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \tag{62}$$

$$q_0^y = \frac{1}{\sqrt{3}}\left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m\right), \tag{63}$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2}\begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad \text{(k = m =0)} \tag{64}$$

$$\vec{K}' = \frac{\pi}{2}\begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad \text{(k = -1, m =0).} \tag{65}$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners.

0.1.4 small momentum expansion

To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

$$f\left(\vec{q}_0 + \delta\vec{q}\right) \approx f(\vec{q}_0) + \left(\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}\right) \cdot \delta\vec{q}. \tag{66}$$

Around $\vec{K}$:

$$\cos\left(\vec{\delta}_1(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_1\vec{K}\right) - \sin\left(\vec{\delta}_1\vec{K}\right)\vec{\delta}_1\vec{p} = -\vec{\delta}_1\vec{p} \tag{67}$$

$$\sin\left(\vec{\delta}_2(\vec{K}+\vec{p})\right) \approx \sin\left(\vec{\delta}_2\vec{K}\right) + \cos\left(\vec{\delta}_2\vec{K}\right)\vec{\delta}_2\vec{p} = \vec{\delta}_2\vec{p} \tag{68}$$

$$\cos\left(\vec{\delta}_3(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_3\vec{K}\right) - \sin\left(\vec{\delta}_3\vec{K}\right)\vec{\delta}_3\vec{p} = -\vec{\delta}_3\vec{p} \tag{69}$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \tag{70}$$

$$\vec{\delta}_2\vec{K} = 0 \tag{71}$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \tag{72}$$

to obtain

$$\Rightarrow H_{\vec{K}} \approx 2t\sum_{p,\sigma}\begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} \end{pmatrix}\begin{pmatrix} -\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix}\begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{73}$$

And around $\vec{K}' = -\vec{K}$:

$$\Rightarrow H_{\vec{K}'} \approx 2t\sum_{p,\sigma}\begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} \end{pmatrix}\begin{pmatrix} \vec{\delta}_1\vec{p} & \left(\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix}\begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{74}$$

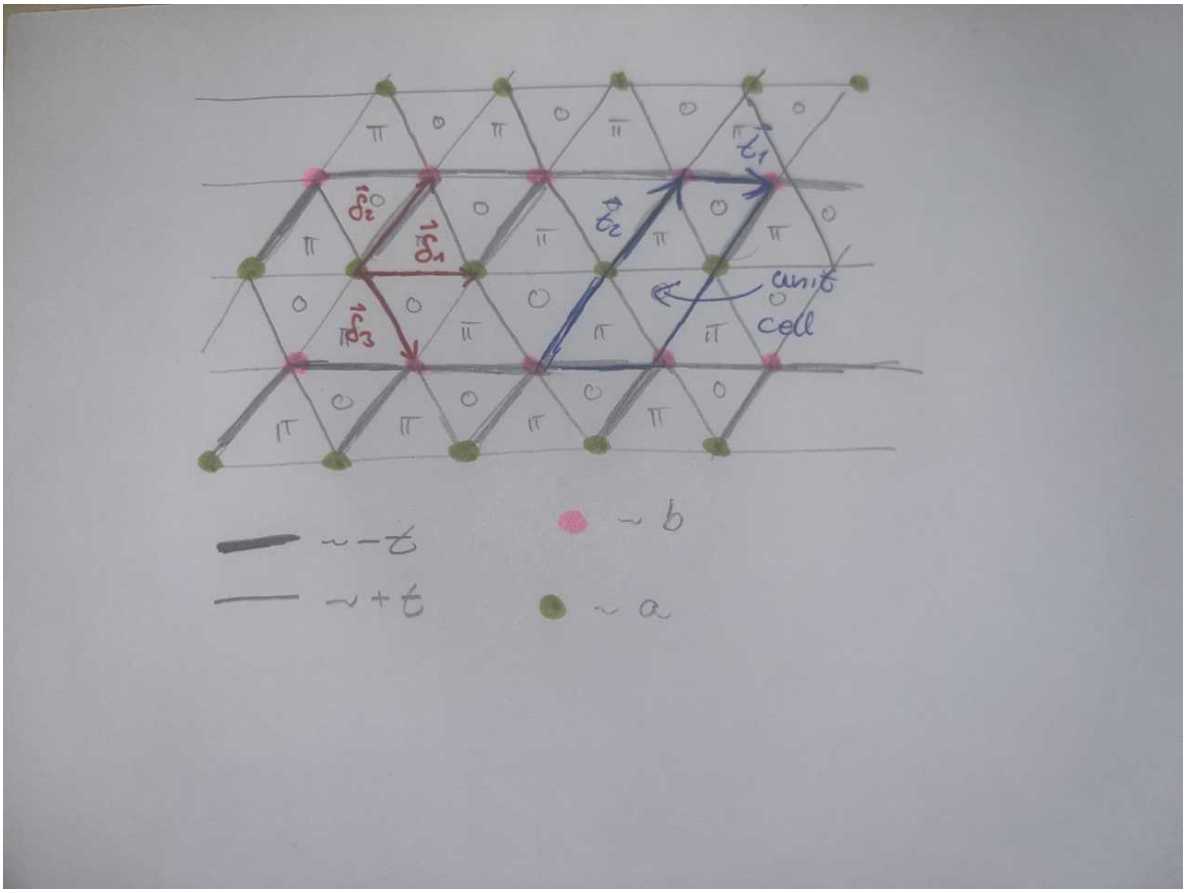
This gives the dispersion relation (for both cones)

$$\begin{aligned} \varepsilon &\approx 2t\sqrt{p_x^2 + \frac{1}{4}\left(p_x + \sqrt{3}p_y\right)^2 + \frac{1}{4}\left(p_x - \sqrt{3}p_y\right)^2} \\ &= \sqrt{6}t\sqrt{p_x^2 + p_y^2} \\ &= \sqrt{6}t|\vec{p}| \end{aligned} \tag{75}$$

Thus, the fermi velocity is $v_F = \sqrt{6}t$. This result has been reported before here [5]. $\vec{K}$ and $\vec{K}'$ are time revearsel partners, thus the two hamiltonains should map onto each other under time revearesal. I would guess that $T = -C$ in k-space, where C is complex conjugation, which would be consistent with the form of the Hamiltonian.

0.1.5 Continuum Model/ Gradient Expansion

Next I will derive the low-energy Hamilotnian in real space, as a continuum theory (/field theory). I start by writing down the real-space hamiltonian, I reshuffled term a bit compared to when I first wrote it down above, the chemical potentials are set to zero.



$$H = \frac{t}{2}\sum_{i\sigma}\Big(a^\dagger_{i\sigma}a_{i+\delta_1;\sigma} - a^\dagger_{i\sigma}b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma}b_{i+\delta_3;\sigma} - b^\dagger_{i\sigma}b_{i+\delta_1;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_2;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_3;\sigma} + \text{h.c.}\Big) \tag{76}$$

I chose a bad method for indexing the lattice position, I should eventually revise this.

The idea is to decompose the the real-space wavefuntion by its momentum - one part is the momentum is the momentum of the dirac point, the other point describes a small deviation from the dirac point (small wavevector -¿ large wavelength -¿ slowly varying in real space). This justifies a gradient expansin of the "deviation field". think of a better explenation for this. But I guess its correct in its spirit.

$$\begin{aligned} a_{i\sigma} &\sim e^{i\vec{K}\vec{R}_i}A_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}A_{K'\sigma}(\vec{R}_i) \\ b_i &\sim e^{i\vec{K}\vec{R}_i}B_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}B_{K'\sigma}(\vec{R}_i) \end{aligned} \tag{77}$$

$A_{K\sigma}, A_{K'\sigma}, B_{K\sigma}$ aqnd $B_{K'\sigma}$ are slowly varying fermionic fields. I begin with the expansion around the K-point. If the above expansion is correct, the cross terms should vanish. Show this. The protoypical method is

$$
\begin{aligned}
a^\dagger_{i\sigma}a_{i+\delta;\sigma} &\sim e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)A_{K\sigma}(\vec{R}_i+\vec{\delta}) \\
&\approx e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)\left(1+\vec{\delta}\vec{\nabla}_{\vec{R}}\right)A_{K\sigma}(\vec{R}_i) \\
&= e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)\left(1+i\vec{\delta}\vec{p}\right)A_{k\sigma}(\vec{R}_i),
\end{aligned} \tag{78}
$$

where $\vec{p}=-i\vec{\nabla}_{\vec{R}}$. The diagonal terms are

$$
A^\dagger_{K\sigma}(\vec{R}_i)\left[e^{i\vec{K}\vec{\delta}_1}(1+i\vec{\delta}_1\vec{p})+e^{-i\vec{K}\vec{\delta}_1}(1-i\vec{\delta}_1\vec{p})\right]A_{K\sigma}(\vec{R}_i) \tag{79}
$$

$$
B^\dagger_{K\sigma}(\vec{R}_i)\left[-e^{i\vec{K}\vec{\delta}_1}(1+i\vec{\delta}_1\vec{p})-e^{-i\vec{K}\vec{\delta}_1}(1-i\vec{\delta}_1\vec{p})\right]B_{K\sigma}(\vec{R}_i). \tag{80}
$$

The off-diagonal terms are

$$
A^\dagger_{K\sigma}(\vec{R}_i)\frac{t}{2}\left[-e^{i\vec{\delta}_2\vec{K}}(1+i\vec{\delta}_2\vec{p})+e^{-i\vec{\delta}_2\vec{K}}(1-i\vec{\delta}_2\vec{p})+e^{i\vec{\delta}_3\vec{K}}(1+i\vec{\delta}_3\vec{p})+e^{-i\vec{\delta}_3\vec{K}}(1-i\vec{\delta}_3\vec{p})\right]B_{K\sigma}(\vec{R}_i) \tag{81}
$$

$$
B^\dagger_{K\sigma}(\vec{R}_i)\frac{t}{2}\left[e^{i\vec{\delta}_2\vec{K}}(1+i\vec{\delta}_2\vec{p})-e^{-i\vec{\delta}_2\vec{K}}(1-i\vec{\delta}_2\vec{p})+e^{i\vec{\delta}_3\vec{K}}(1+i\vec{\delta}_3\vec{p})+e^{-i\vec{\delta}_3\vec{K}}(1-i\vec{\delta}_3\vec{p})\right]A_{K\sigma}(\vec{R}_i) \tag{82}
$$

Use

$$
\vec{\delta}_1\vec{K}=\frac{\pi}{2} \tag{83}
$$

$$
\vec{\delta}_2\vec{K}=0 \tag{84}
$$

$$
\vec{\delta}_3\vec{K}=\frac{\pi}{2} \tag{85}
$$

to obtain

$$
H_{\vec{K}}\approx\sum_\sigma 2t\int d^2\vec{R}\quad\Psi^\dagger_{K\sigma}(\vec{R})\begin{pmatrix}-\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3-i\vec{\delta}_2\right)\vec{p}\\ \left(-\vec{\delta}_3+i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p}\end{pmatrix}\Psi_{K\sigma}(\vec{R}), \tag{86}
$$

with

$$
\Psi_{K\sigma}(\vec{R})=\begin{pmatrix}A_{K\sigma}(\vec{R})\\ B_{K\sigma}(\vec{R})\end{pmatrix}. \tag{87}
$$

Doing the same around $\vec{K}'$ gives

$$
H_{\vec{K}'}\approx\sum_\sigma 2t\int d^2\vec{R}\quad\Psi^\dagger_{K'\sigma}(\vec{R})\begin{pmatrix}\vec{\delta}_1\vec{p} & \left(\vec{\delta}_3-i\vec{\delta}_2\right)\vec{p}\\ \left(\vec{\delta}_3+i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p}\end{pmatrix}\Psi_{K'\sigma}(\vec{R}) \tag{88}
$$

Maybe it is handy to write $\vec{\delta}_1=\vec{\delta}_2+\vec{\delta}_3$. When Comparing this to the results obtained by expanding in reciprocal space, keep in mind that here $\vec{p}$ is an operator. So technically, one would have to do a kind of fourier transform for the fermionic fields to find the exact representation as found in recriprocal space. This is why the Hamiltonian looks so similar, even though we are still in real space here.

Imagine we did this small frequency? Fourier expansion and we found the reciprocal space Hamiltonian (replace the momentum operator by its eigenvalue!), the corresponding disperion relation would then read

$$\varepsilon(\vec{p}) = \left| 2t \begin{pmatrix} \mp\vec{\delta}_3\vec{p} \\ \vec{\delta}_2\vec{p} \\ \mp\vec{\delta}_1\vec{p} \end{pmatrix} \right| = \sqrt{6}t\sqrt{p_x^2 + p_y^2} = \sqrt{6}t|\vec{p}|, \tag{89}$$

which complies with the k-space result. In case I want to diagonalize this in operator form at some point; since the problem is non-interacting there must be a harmonic oscillator hidden in there.

0.2 0-π State on Strained Lattice

references are [3], [2], [6], [1] and [7]

0.2.1 Basics of Strain

For the purpose of introducing strain into the model, I first establish its effect on the coordinates and exchange couplings

$$\vec{R}_i \mapsto \vec{R}'_i = \vec{R}_i + \vec{U}(\vec{R}_i) \tag{90}$$
$$\vec{\delta}_{ij} \mapsto \vec{\delta}'_{ij} = \vec{\delta}_{ij} + \vec{U}(\vec{R}_j) - \vec{U}(\vec{R}_i) \tag{91}$$

$\vec{\delta}_{ij} = \vec{R}_j - \vec{R}_i$;)

$\vec{U}(\vec{R})$ is the displacement field. We approximate that $\vec{U}$ varies slowly with $\vec{R}$:

$$\vec{U}(\vec{R}_j) = \vec{U}(\vec{R}_i + \vec{\delta}_{ij}) \approx \vec{U}(\vec{R}_i) + \left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} \tag{92}$$
$$\tag{93}$$

It is convenient to redefine

$$\left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} = \underline{u}\cdot\vec{\delta}_{ij}, \tag{94}$$

where $u_{\alpha\beta} = \partial_\beta U_\alpha$ is the strain tensor.

$$\Rightarrow \vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{95}$$

Here $\underline{u}$ is taken at $\vec{R}_i$, however one can derive the same expression but with $\underline{u}$ taken at $\vec{R}_j$. In what follows I choose the expansion around $\vec{R}_i$ as shown in the main text. This however is arbitrary. Something dependening on $\vec{R}_i$ and $\vec{R}_j$ (or $(\vec{R}_i + \vec{R}_j)/2$) would seem more sensible but would complicate later expressions. I was not able to resolve this yet.

I do the same approximation for the exchange couplings

$$J(\vec{\delta}_{ij}) \mapsto J(\vec{\delta}'_{ij}) \approx J\left(\vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij}\right) \tag{96}$$
$$\approx J(\vec{\delta}_{ij}) + \left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\left(\vec{\nabla}_{\vec{\delta}}J(\vec{\delta})\big|_{\delta_{ij}}\right) \tag{97}$$

For the function $J(\vec{\delta})$ I uses the form

$$J(\vec{\delta}) = J\exp\left[-\beta\left(|\vec{\delta}|/a - 1\right)\right] \approx J\left(1 - \beta\left(|\vec{\delta}|/a - 1\right)\right), \tag{98}$$

, see e.g. [4], where a is the lattice constant and $\beta = -d\ln(J)/d\ln\left(|\vec{\delta}|\right)\big|_a$ the magneto-elastic coupling.

$$\Rightarrow \vec{\nabla}_{\vec{\delta}} J(\vec{\delta})\big|_{\delta_{ij}} \overset{|\vec{\delta}_{ij}|=a}{=} -\frac{J\beta}{a^2}\vec{\delta}_{ij}$$
$$\Rightarrow J(\vec{\delta}_{ij}) \approx J(\vec{\delta}_{ij}) - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{99}$$

Looking at the matrix elments (79 and below), one also requires the expansion of $\exp\left[i\vec{\delta}_{ij}\vec{K}\right]$. To summarize:

$$\vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{100}$$
$$J(\vec{\delta}'_{ij}) \approx J - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{101}$$
$$e^{i\vec{K}\cdot\vec{\delta}'_{ij}} \approx e^{i\vec{K}\cdot\vec{\delta}_{ij}} + ie^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{102}$$

In this approximation the Mean field does not react to the strain at all??

0.2.2 Straining the 0-π-State

The prototypical form of the matrix elements is (expanding around $\vec{K}$)

$$J(\vec{\delta}_{ij})a^\dagger_{i\sigma}a_{i+\delta_{ij};\sigma} \sim A^\dagger_{K\sigma}\left(J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p})\right)A_{K\sigma} \tag{103}$$

Expanding up to first order in strain I acquire the following terms

$Je^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p})$	unstrained Hamiltonian/ zeroth order
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$-i\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\vec{p}$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$

(I drop the lattice constant again from now on) I will focus on the terms of order zero + terms of either order 1 in field gradient or order 1 in strain [3]. I havent come up with a good argument for this yet. How does one even compares orders between different expansions? This gives

$$\begin{aligned} J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p}) &\sim Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\vec{p}+i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \\ &= Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\left[\vec{p}+\underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right)\right]\right) \end{aligned} \tag{104}$$

Hence, on this level

$$\vec{A} = \underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right). \tag{105}$$

This is not correct since one in has to add contributions from different local directions in the off-diagonal terms of H!

$$B = (\partial_x A_y - \partial_y A_x) \tag{106}$$

It turns out however, that $\vec{B} = 0$, which is explained in what follows. The first term is simply rotationless (this was shown here [6]) and can therefore be "gauged away".

Regarding the second term; I take a step back to the dispersion relation of the unstrained model

$$H = \sum_{\vec{q}\sigma} \Psi^{\dagger}_{q\sigma} \cdot \left(\vec{h}(\vec{q}) \cdot \vec{\sigma}\right) \cdot \Psi_{q\sigma} \quad \text{with} \quad \vec{h}(\vec{q}) = 2t \begin{pmatrix} \sin\left(\vec{q}\cdot\vec{\delta}_3\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_1\right) \end{pmatrix} \Rightarrow \varepsilon(\vec{q}) = \left|\vec{h}(\vec{q})\right|. \tag{107}$$

By definition, $\varepsilon(\vec{K}^{(\prime)}) = 0$. For this to be true, every component of $\vec{h}$ must vanish individually at $\vec{K}^{(\prime)}$. This gives three conditions for $\vec{q}$, one for each direction.

Evidently each expression in $\vec{h}$ is obtained by adding contribtions from hopping forwards and backwards along $\vec{\delta}_{ij}$. However, since $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ is invariant under $\vec{\delta}_{ij} \mapsto -\vec{\delta}_{ij}$ it acquires the same dispersion-like prefactor as the zeroth order, which vanishes by definition at $\vec{K}^{(\prime)}$.

So why does the term vanish? Its a mix of the special structure of the hamiltonian (i.e. the local directions decouple in the pauli-matrix basis) and the fact that the expansion term $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ cant distinguish between $\vec{\delta}_{ij}$ and $-\vec{\delta}_{ij}$. The latter statement is obviously not model-specific. The same mechanism shouild be present in the π-flux square lattice. For uniform hopping (e.g. ordinary Graphene) it is absent (the decoupling of the directions is a cosnequence of the non-uniform hopping/ the mean-field).

It is not clear to me how this can be upheld under gauge transformations (wouldn't this mix directions in the Pauli-matrix basis?)

I would expect to find a pseudo-magnetic field in higher orders of strain?

0.3 Strain Engineering and other Games with the 0-π-State

0.4 Strain games

In this section I try

References

[1] A. H. Castro Neto et al. "The electronic properties of graphene". In: *Rev. Mod. Phys.* 81 (1 Jan. 2009), pp. 109–162. DOI: 10.1103/RevModPhys.81.109. URL: https://link.aps.org/doi/10.1103/RevModPhys.81.109.

[2] Alexander L. Kitt et al. "Lattice-corrected strain-induced vector potentials in graphene". In: *Phys. Rev. B* 85 (11 Mar. 2012), p. 115432. DOI: 10.1103/PhysRevB.85.115432. URL: https://link.aps.org/doi/10.1103/PhysRevB.85.115432.

[3] Mary Madelynn Nayga, Stephan Rachel, and Matthias Vojta. "Magnon Landau Levels and Emergent Supersymmetry in Strained Antiferromagnets". In: *Phys. Rev. Lett.* 123 (20 Nov. 2019), p. 207204. DOI: 10.1103/PhysRevLett.123.207204. URL: https://link.aps.org/doi/10.1103/PhysRevLett.123.207204.

[4] Vitor M. Pereira, A. H. Castro Neto, and N. M. R. Peres. "Tight-binding approach to uniaxial strain in graphene". In: *Phys. Rev. B* 80 (4 July 2009), p. 045401. DOI: 10.1103/PhysRevB.80.045401. URL: https://link.aps.org/doi/10.1103/PhysRevB.80.045401.

[5] Stephan Rachel et al. "Quantum Paramagnet in a π Flux Triangular Lattice Hubbard Model". In: *Phys. Rev. Lett.* 114 (16 Apr. 2015), p. 167201. DOI: 10.1103/PhysRevLett.114.167201. URL: https://link.aps.org/doi/10.1103/PhysRevLett.114.167201.

[6] M. Ramezani Masir, D. Moldovan, and F.M. Peeters. "Pseudo magnetic field in strained graphene: Revisited". In: *Solid State Communications* 175-176 (Dec. 2013), pp. 76–82. ISSN: 0038-1098. DOI: 10.1016/j.ssc.2013.04.001. URL: http://dx.doi.org/10.1016/j.ssc.2013.04.001.

[7] Junsong Sun et al. "Pseudomagnetic fields in square lattices". In: *Phys. Rev. B* 108 (20 Nov. 2023), p. 205149. DOI: 10.1103/PhysRevB.108.205149. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.205149.